

With personal computing and TCP/IP in place by 1989, the '90s had it easy

based personal computer. Running on an MC68000 CPU by Motorola at 8 MHz, it had 128 KB of RAM—decent for those days, but not too much. The OS was, of course, Mac OS 1.0. While the Mac popularised the GUI, it didn't do too much for games: the non-console gaming honours for the decade go to the little monitor-less, floppy-less microcomputers of the time—notably the Commodore 64, BBC Micro, and ZX Spectrum. In retrospect, it's absolutely amazing how much game developers packed into 48 K and 64 K those days! In fact, these computers, while running BASIC compilers and all, were primarily used as gaming devices, being hooked up to a TV set. Games were loaded via tape—it took about five minutes for one game to load—and some folks bought drives such as the ZX Microdrive, which loaded games via cartridges much faster. (But those of us who were there remember that tape was the best way to go because they could be easily replicated!)

The beginning of the decade also witnessed the optical storage revolution: the Philips-Sony-CD story. In 1979, Sony and Philips set up a joint engineering task force with the mandate of designing a standardised digital audio disc out of the ashes of earlier efforts at promoting the optical disc. (If you remember, the first CD had already been invented in the '60s by James Russell.) This task force soon came up with the "Red Book"—the CD standard. The disc based on that standard soon reached the market—in 1982 in Asia.

There's an apocryphal story about the maximum length of the audio CD being set at 74 minutes so it could hold a recording of Beethoven's Ninth Symphony. Some sources say the reason was because it was Sony president

Norio Ohga's favourite piece of music. Honestly, we can't vouch for the veracity of this story—or of any others. Anyway, Billy Joel's *52nd Street* was the first CD released commercially, not a recording of the Ninth Symphony.

By 1984, CDs were household things (at least in developed countries). In 1985, evolved out of the music CD, the CD-ROM drive stepped out to the market.

The first IBM PC clone came out in 1982. It was by a company with the sedate name of Columbia Data Products; Compaq soon came out with their own version. The x86 revolution essentially happened in the '80s: 1984 saw IBM releasing the PC Jr. and the PC-AT. The former flopped, the latter didn't: it was based around Intel's 80286 processor, and became popular. Later on, Compaq released the obviously-named Deskpro 386, the first PC to use the 80386 processor from Intel.

The 386 was 32-bit, and computers based on the 386 were as powerful as some mainframes of yore. The 486 debuted in 1989, but it wasn't as much of a landmark—with the 386, "modern" desktop computing was already in place.

Windows, too, actually has its origins in the '80s (and not in the '90s), with Windows 1.0 having been developed in 1985. Was it just a front-end for DOS? Many believe so, but there are differences: Windows EXEs had a file format called "new executable," which only Windows could process. There was also Windows' own memory management system, which implemented a virtual memory scheme. Still, Windows 1.0 didn't become hugely popular. Version 2.0 enjoyed more success, and from then on, it's a history lesson.

Meanwhile, the decade saw the introduction of commercial e-mail. But the most important develop-

ment here was inarguably TCP/IP: in 1982, DCA (the Defense Communications Agency) and ARPA established TCP/IP as the protocol suite for ARPANET. The definition of the word "internet" was set: "an internet" was now defined as a connected set of networks running on TCP/IP, and "The Internet" was defined as the set of connected TCP/IP internets. And so, the Internet had arrived: the first name server was developed at the University of Wisconsin, and the DNS system was introduced by 1984. And in 1986, the first domain name was registered. Symbolics.com. Interestingly, Symbolics.com is today a plain HTML page!

With personal technology—including personal computing—and TCP/IP in place, the '90s had it easy. The x86 architecture simply had to improve upon itself in accordance with Moore's Law of 1965, Windows would have to become Windows 95, and the Internet would have to grow a million-fold—but that was just a matter of hooking up computers to the phone system. On to a world we're familiar with... ■

IBM PC, x86, Windows

The star of the '80s was obviously the personal computer, but Apple or PC? Motorola or x86? It's a tough choice and a touchy subject. Many reasons are cited for why the Mac is not our PC today: it was underpowered for the GUI it sported, and DOS was faster. Macs were more expensive than PCs. And perhaps most important is the fact that Apple didn't allow a Mac clone industry (except for a brief time, and to disastrous effect, to be historically accurate)—the world might well have been different if IBM patented the PC architecture and kept it proprietary.

Yes, the IBM PC is the best thing the 1980s gave us. x86 is a phenomenon in itself: IBM's decision to use the architecture was criticised, and the Motorola 68000 was considered a much better alternative. But due to certain accidents of history, x86 it was.

And then there was the rise of Windows. Apple had already become niche by the early '90s.

